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CS & DA

Probability and Statistics

DPP: 01

Joint PMF & PDF

Q1 Consider two random variables X and Y with joint PMF given in Table.

	Y= 2	Y= 4	Y= 5
X=1	1/12	1/24	1/24
X= 2	1/6	1/12	1/8
x = 3	1/4	1/8	1/12

1) Find P (X $\leq$2, Y $\leq$4).
2) Find the marginal PMF's of X and Y.
3) Find P (Y =2| X = 1).
4) Are X and Y independent?

Q2 For the adjoining bivariate probability distribution of X and Y, find:
(i) P(x ≤ 1, Y = 2),
(ii) P(X ≤ 1),
(iii) P(Y ≤ 3), and
(iv) P(X < 3, Y ≤ 4).

y x	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Q3 For the joint probability distribution of two random variable X and Y given below:

Y X	1	2	3	4	Total
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find 1) the marginal distributions of X and Y, and 2) conditional distribution of X given the value of Y = 1 and that of Y given the value of X = 2.

Q4 I have a bag containing 40 blue marbles and 60 red marbles. I choose 10 marbles (without replacement) at random. Let X be the number of blue marbles and y be the number of red marbles. Find the joint PMF of X and Y.

Q5 The joint p.d.f of two random variable X and Y is given by :

$$f(x,y) = \frac{9(1+x+y)}{2(1+x)^4(1+y)^4}; \begin{pmatrix} 0 \leq x < \infty \\ 0 \leq y < \infty \end{pmatrix}$$

Find the marginal distribution of X and Y, and the conditional distribution of Y for X = x.

Q6 Given:
$f(x,y) = e^{-(x+y)}; 0 \leq x < \infty, 0 \leq y < \infty$
find
1)P(X > 1)
2) P (X < Y | X < 2Y)
3) P (1 < X + Y < 2)

Q7 Let X and Y be jointly continuous random variables with Joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cx+1 & x,y \geq 0, x+y<1 \\ 0 & otherwise \end{cases}$$

1) Find the constant c.
2) Find the marginal PDFs $f_x(x)$ and $f_Y(y)$.
3) Find P(Y<2X^2).

Q8 Let X and Y be jonitly continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)} & x,y \geq 0 \\ 0 & otherwise \end{cases}$$

1) Are X and Y independent?
2) Find E[Y|X>2].
3) Find P (X > Y).

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Q9 Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

We know that given X = x , the random variable Y is uniformly distributed on [−x, x].

1) Find the jont PDf F_{XY} (x,y).

2) Find f_Y(y).

Q10 A bank operates a drive-up and a walk-up window.Here

X= the proportion of time the drive-up facility is in use.

Y= the proportion of time the walk-up window in use.

$$f(x,y) = \begin{cases} \frac{6}{5}\left(x+y^2\right) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & otherwise \end{cases}$$

The probability that neither facility is busy more than one-quarter of the time is?

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Answer Key

Q1 $\frac{3}{8}$, 1/2

Q2 1/16,28/32,23/64,9/16

Q3 Refer to hint for solution

Q4 Refer to hint for sol

Q5 Refer to the hint for solution

Q6 $e^{-1}, \frac{3}{4}, \frac{2}{e} - \frac{3}{e^2}$

Q7 3,53/96

Q8 $1) f_X(x) = 2e^{-2x}, x \geq 0 \; and \; f_Y(y) = 3e^{-3y}, y \geq 0$
$2) \frac{1}{3}$
$3) \frac{3}{5}$

Q9 $f_{XY}(x, y) \begin{cases} 1 & |y| \leq x \leq 1 \\ 0 & otherwise \end{cases}$

Q10 7/640

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Hints & Solutions

Q1 Text Solution:

1)

To find P (X $\leq$2, Y $\leq$4), we can write

P (X $\leq$2, Y $\leq$4) = P_{XY} (1, 2) + P_{XY} (1, 4) + P_{XY} (2, 2) + P_{XY} (2, 4)

$$= \frac{1}{12} + \frac{1}{24} + \frac{1}{6} + \frac{1}{12} = \frac{3}{8}.$$

2)

Note from the table that

R_X {1,2,3} and R_Y = {2,4,5}

$$.P_X(y) = \begin{cases} \frac{1}{2} & y = 2 \\ \frac{1}{4} & y = 4 \\ \frac{1}{4} & y = 5 \\ 0 & \text{otherwise} \end{cases}$$

$$P_X(x) = \begin{cases} \frac{1}{6} & x = 1 \\ \frac{3}{8} & x = 2 \\ \frac{11}{12} & x = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$P_X(y) = \begin{cases} \frac{1}{2} & y = 2 \\ \frac{1}{4} & y = 4 \\ \frac{1}{4} & x = 5 \end{cases}$$

3) Using the formula for conditional probability, we have

$$P(Y = 2|X = 1) = \frac{P(X=1 \cap Y=2)}{P(X=1)}$$

$$= \frac{P_{XY}(1,2)}{P_X(1)}$$

$$= \frac{\frac{1}{12}}{\frac{1}{6}} = \frac{1}{2}$$

4)

P(X=x_i ∩Y = y_i) is equal to P(X=x_i).(P(Y = y_i))

$$P(X = 2, Y = 2) = \frac{1}{6}$$

$$\neq P(X = 2)P(Y = 2) = \frac{3}{16}.$$

Thus, we conclude that X and Y are not independent.

Q2 Text Solution:

The marginal distributions are given below:

y / x	1	2	3	4	5	6	P_X(x)
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
P_y(Y)	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	ΣP(x) = 1 ΣP(y) = 1

(i) P(x ≤ 1, Y = 2) = P(X = 0,Y = 2) + P(X = 1,Y = 2) = 0 + $\frac{1}{16} = \frac{1}{16}$

(ii) P(X ≤ 1) = P(X = 0) + P(X = 1) = $\frac{8}{32} + \frac{10}{16} = \frac{7}{8}$

(iii) P(Y ≤ 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = $\frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$

(iv) P(X < 3, Y ≤ 4) = P(X = 0, Y ≤ 4) + P(X = 1, Y ≤ 4) + P (X = 2, Y ≤ 4)

$$= \left(\frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right) = \frac{9}{16}$$

Q3 Text Solution:

(i) The marginal distributions of X is defined as:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$P(X = 1) = \sum_y P(X = 1, Y = y)$$

= P(X = 1, Y = 1) + P (X = 1, Y = 2) + P(X = 1, Y = 3) + P (X = 1, Y = 4)

$$= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$$

Similarly,

$$P(X = 2) = \sum_y P(X = 2, Y = y) = \frac{9}{36};$$

$$P(X = 3) = \sum_y P(X = 3, Y = y) = \frac{8}{36}$$

$$P(X = 4) = \sum_y P(X = 4, Y = y) = \frac{9}{36}.$$

Similarly, we can obtain the marginal distribution of Y.

Value of X, x	1	2	3	4
P (X = x)	$\frac{1}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

Value of Y, y	1	2	3	4
P (Y = y)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

(ii) The conditional probability function of X given Y is defined as follows:

$$P(X = x|Y = y) = \frac{P(X=x|Y=y)}{P(Y=y)}.$$ Therefore

$$P(X = 1|Y = 1) = \frac{P(X=1|Y=1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$

$$P(X = 2|Y = 1) = \frac{P(X=2|Y=1)}{P(Y=1)} = \frac{1/36}{11/36} = \frac{1}{11}$$

$$P(X = 3|Y = 1) = \frac{P(X=3|Y=1)}{P(Y=1)} = \frac{5/36}{11/36} = \frac{5}{11}$$

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$P(X=4|Y=1)=\frac{P(X=4|Y=1)}{P(Y=1)}=\frac{1/36}{11/36}=\frac{1}{11}$

Hence the conditional distribution of X given Y = 1 is:

x :	1	2	3	4
P(X = x | Y = 1)	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{4}{11}$

Similarly, we can obtain the condition of Y for X = 2 as given below:

y :	1	2	3	4
P(Y = y | X = 2)	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{9}$

Q4 Text Solution:

This is, in fact a hypergeometic distribution. First note that we must have X+Y = 10, So

P_{XY} ={(i, j)| i + j = 10, i , j $\in \mathbb{Z}$, i ,j$\geq$ 0}

= {(0, 10), (1, 9) , (2, 8), ..., (10, 0)}.

P_{XY}(i, j) =

$$\begin{cases} \frac{\binom{40}{i}\binom{60}{j}}{\binom{100}{10}} & i+j=10, i,j\in\mathbb{Z}, i,j\geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Q5 Text Solution:

Marginal p.d.f of X is given by

$f_x(x)=\int_0^\infty f(x^4,y)\,dy$

$=\frac{9}{2(1+x)^4}\cdot\int_0^\infty\frac{(1+y)+x}{(1+y)^4}+dy$

$=\frac{9}{2(1+x)^4}\cdot\int_0^\infty\left[(1+y)^{-3}+x(1+y)^{-4}\right]dy$

$=\frac{9}{2(1+x)^4}\cdot\left[\left|\frac{-1}{2(1+y)^2}\right|_0^\infty+x\left|\frac{-1}{3(1+y)^3}\right|_0^\infty\right]$

$=\frac{9}{2(1+x)^4}\cdot\left[\frac{1}{2}+\frac{x}{3}\right]$

$=\frac{3}{4}\cdot\frac{3+2x}{(1+x)^4};0\leq x<\infty$

Since f(x, y) is symmetric in x and y, the marginal p.d.f of y is given by

$f_r(y)=\int_0^\infty f(x,y)\,dx$

$=\frac{3}{4}\cdot\frac{3+2y}{(1+y)^4};0\leq y<\infty$

The conditional distribution of Y for X = x is given by

$f_{xy}(Y=y|X=x)=\frac{f_{xy}(x,y)}{f_x(x)}$

$=\frac{9(1+x+y)}{2(1+x)^4(1+y)^4},\frac{4(1+x)^4}{3(3+2x)}$

$=\frac{6(1+x+y)}{(1+y)^4(3+2x)};0\leq y<\infty$

Q6 Text Solution:

We are given:

$f(x,y)=e^{-(x+y)};0\leq x<\infty,0\leq y<\infty$ ....(1)

= (e^{-x}) (e^{-y})

$=f_x(x)\cdot f_y(y);0\leq x<\infty,0\leq y<\infty$

⇒ X and Y are independent and

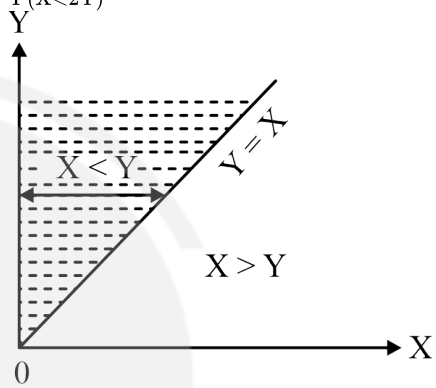
$f_x(x)=e^{-x};x\geq 0$ and $f_y(y)=e^{-y};y\geq 0$ ....(2)

1) $P(X>1)=\int_1^\infty f_x(x)dx=\int_1^\infty e^{-x}\,dx$

$=\left|\frac{e^{-x}}{-1}\right|_l^\infty=\frac{1}{e}$

2) $P(X<Y|X<2Y)=\frac{P(X<Y\cap X<2Y)}{P(X<2Y)}$

$\frac{P(X<Y)}{P(X<2Y)}$ .... (3)



$P(X<Y)=\int_0^\infty\left[\int_0^Y f(x,y)\,dx\right]dy$

$=\int_0^\infty\left[e^{-y}\left|\frac{e^{-x}}{-1}\right|_0^y\right]dy=-\int_0^\infty e^{-y}(e^{-y}-1)\,dy$

$=\left|\frac{e^{-2y}}{-2}+e^{-y}\right|_0^\infty=1-\frac{1}{2}=\frac{1}{2}$

$P(X<2Y)=\int_0^\infty\left[\int_0^{2y}f(x,y)\,dx\right]dy=-$

$\int_0^\infty e^{-y}(e^{-2y}-1)\,dy$

$=-\left|\frac{e^{-3y}}{-3}+e^{-y}\right|_0^\infty=1-\frac{1}{3}=\frac{2}{3}$

Substituting in (3)

$P(X<Y|X<2Y)=\frac{1/2}{2/3}=\frac{3}{4}$

3) $P(1<X+Y<2)=\iint_1 f(x,y)dxdy=$

$\iint_{11} f(x,y)\,dxdy$

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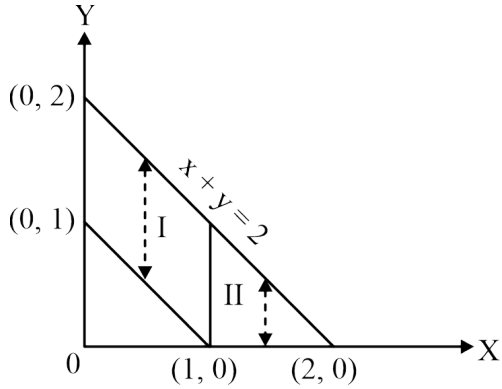

Random Variables-Distribution Functions

$= \int_0^1 \left[\int_{1-x}^{2-x} f(x,y)\,dy\right] dx +$

$\int_1^2 \left[\int_0^{2-x} f(x,y)\,dy\right] dx$

$= \int_0^1 \left[e^{-x} \int_{1-x}^{2-x} e^{-y}\,dy\right] dx +$

$\int_1^2 \left[e^{-x} \int_0^{2-x} e^{-y}\,dy\right] dx$

$= = \int_0^1 \left[\frac{e^{-x}}{-1}\left(e^{x-2} - e^{x-1}\right)\right] dx +$

$\int_1^2 \left[\frac{e^{-x}}{-1}\left(e^{x-2} - 1\right)\right] dx$

$= -\left(e^{-2} - e^{-1}\right) \int_0^1 1 \cdot dx - \int_1^2 \left(e^{-2} - e^{-x}\right) dx$

$= -\left(e^{-2} - e^{-1}\right) |x|_0^1 - \left|e^{-2} \cdot x + e^{-x}\right|_1^2$

$= \frac{2}{e} - \frac{3}{e^2}$

Q7 Text Solution:

To find the constant c, we write

$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y)dxdy$

$= \int_0^1 \int_0^{1-x} cx + 1\, dydx$

$= \int_0^1 (cx+1)(1-x)dx$

$= \frac{1}{2} + \frac{1}{6}c.$

Thus, we conclude c = 3.

2) We first not that $R_X = R_Y = [0,1]$

$fx(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy$

$= \int_{-\infty}^{1-x} 3x + 1\, dy$

= (3x +1) (1−x) , for x [0,1]

Thus, we have

$f_X(x)$

$= \begin{cases} (3x+1)(1-x) & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$

Similarly, we obtain

$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx$

$= \int_0^{1-y} 3x + 1\, dx$

$= \frac{1}{2}(1-y)(5-3y), for\ y \in [0,1].$

Thus, we have

$f_Y(y)$

$= \begin{cases} \frac{1}{2}(1-y)(5-3y) & 0 \le y \le 1 \\ 0 & otherwise \end{cases}$

3) To find $P(Y < 2X^2)$ we need to integrate $f_{XY}(x,y)$ over the region shown in figure.

$P\left(Y < 2X^2\right) = \int_{-\infty}^{\infty} f_{XY}(x,y)dydx$

$\int_0^1 \int_0^{\min(2x^2, 1-x)} 3x + 1dydx$

$= \int_0^1 (3x+1)\min\left(2x^2, 1-x\right)dx$

$= \int_0^{\frac{1}{2}} (3x+1)2xdx +$

$\int_{\frac{1}{2}}^1 (3x+1)\left(1-x\right)dx$

$= 53/96$

Q8 Text Solution:

1) In this question we have-

$f_{X,Y}(x,y) = f_X(x)f_Y(y),$

Where

$f_X(x) = 2e^{-2x}, x \ge 0$

$f_Y(y) = 3e^{-3y}, y \ge 0$

Thus they are independent.

2) Since X and Y are independent we have $E[Y|X > 2] = E[Y]$. Note that Y ~Exponential (3) , thus E(Y) = $\frac{1}{3}$.

3)

$P(X > Y) = \int_0^{\infty} \int_y^{\infty} 6e^{-(2x+3y)}dxdy$

$= \int_0^{\infty} 3e^{-5ydy}$

$= 3/5$

Q9 Text Solution:

1) $f_{Y|X}(y|x) = \begin{cases} \frac{1}{2x} & -x \le y \le x \\ 0 & otherwise \end{cases}$

Thus, we have

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$$f\left(x,y\right) = \begin{cases} 1 & 0 \le x \le 1, -x \le y \le x \\ 0 & otherwise \end{cases}$$

$$f\left(x,y\right) = \begin{cases} 1 & |y| \le x \le 1 \\ 0 & otherwise \end{cases}$$

2)

$f_Y\left(y\right) = \int_{-\infty}^{\infty} f_{XY}\left(x,y\right)dx$

$= \int_{|y|}^{1} 1dx$

$= 1 - \left|y\right|.$

Thus,

$$f_Y\left(y\right) = \begin{cases} 1 - |y| & |y| \le 1 \\ 0 & otherwise \end{cases}$$

Q10 Text Solution:

$$f\left(x,y\right) = \begin{cases} \frac{6}{5}\left(x + y^2\right) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & otherwise \end{cases}$$

Say the manager has given us the joint pdf based on his experience:

$p\left(0 < x < \frac{1}{4}, 0 < y < \frac{1}{4}\right) = \int_0^{\frac{1}{4}}$

$\int_0^{\frac{1}{4}} f\left(x,y\right)dxdy$

$\int_0^{\frac{1}{4}} \int_0^{\frac{1}{4}} \frac{6}{5}\left(x + y^2\right)dxdy$

$\frac{6}{5}\left(\int_0^{\frac{1}{4}} \left(\frac{x^2}{2} + xy^2\right)^{\frac{1}{4}} dy\right) = \frac{6}{20}\left(\frac{x^2}{2} + \frac{y^3}{3}\right)_0^{\frac{1}{4}}$

$= \frac{7}{640}$

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